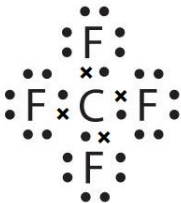


Higher Tier only questions

| Question |     |      |  | Marking details                                                                                                                                                                                                                                            | Marks available |     |     |       |       |      |
|----------|-----|------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                            | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 3        | (a) | (i)  |  | ionic                                                                                                                                                                                                                                                      | 1               |     |     | 1     |       |      |
|          |     | (ii) |  | award (1) for either of following<br><br>melted / made molten<br>dissolved - accept put into water<br><br>neutral answer - heated<br><br>ions are free to move (1)<br>do not accept reference to electrons being free to move                              | 2               |     |     | 2     |       |      |
|          | (b) | (i)  |  | award (2) for correct diagram<br><br><br><br>accept any representation of electrons e.g. all crosses or all dots<br><br>if incorrect award (1) for all four shared pairs |                 | 2   |     | 2     |       |      |